# Project Ideation Matrix

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|---|---|---|---|
| **Date** | July 11, 2026 | **Team ID** | |
| **Project Name** | **India's Agriculture Crop Production Analysis** | | |
| **Phase** | Ideation & Problem Definition | **Maximum Marks** | 2 Marks |

## 1. Executive Project Context

Agriculture remains the backbone of the Indian economy, employing over 50% of the workforce and contributing significantly to the nation's GDP. However, the sector faces extreme volatility due to unpredictable climatic shifts, traditional resource allocation models, and highly fragmented supply chains. This project aims to design a robust data analytics infrastructure that processes historical crop yields, real-time meteorological feeds, soil health datasets, and market pricing patterns. By delivering actionable insights, the platform enables stakeholders to move from reactive mitigation to predictive precision farming.

## 2. Customer Problem Statement Framework

To align development efforts with true stakeholders' requirements, this document utilizes the standardized Customer Problem Statement Framework. This method enforces empathetic alignment, isolating user symptoms from root technical challenges while highlighting the emotional toll of operational inefficiencies.

| | |
|---|---|
| **I am** | **The Customer Archetype:** Describes the specific user profile, demographic attributes, and their direct role within the broader agricultural ecosystem. |
| **I'm trying to** | **The Core Objective / Job-to-be-Done:** Outlines the critical outcome or ultimate business goal that the target user is actively striving to fulfill. |
| **but** | **The Core Operational Friction:** Details the specific systemic barriers, data deficits, or institutional constraints blocking the user's path. |
| **because** | **The Underlying Root Cause:** Probes deep into the technical, structural, or physical constraints that allow the aforementioned friction to persist. |
| **which makes me feel** | **The Emotional Context:** Captures the psychological impact (e.g., risk anxiety, systemic distrust) that damages productivity and delays decision-making. |

## 3. Visual Mapping & Persona Scenarios

Below are conceptual breakdowns mapping specific high-priority target personas to their respective operational constraints within the crop production value chain.

### Scenario A: The Precision Cultivator / Smallholder Farmer

| I AM | I'M TRYING TO | BUT | BECAUSE | WHICH MAKES ME FEEL |
|---|---|---|---|---|
| A smallholder farmer managing regional staple crops with limited capital margins. | Optimize crop yields and secure fair market values during the harvest peak. | Sudden weather extremes disrupt planting cycles, resulting in crop degradation. | Localized weather alerts and soil moisture tracking models are absent or inaccessible. | Highly anxious about financial insolvency and future seasonal planning. |

### Scenario B: The Macro-Level Supply Chain Planner

| I AM | I'M TRYING TO | BUT | BECAUSE | WHICH MAKES ME FEEL |
|---|---|---|---|---|
| A central agricultural storage planner managing national buffer stocks. | Allocate storage capacity efficiently and prevent systemic food waste across states. | Widespread supply bottlenecks occur due to unexpected seasonal yield overflows. | Current predictive models fail to synthesize historical patterns with active acreage data. | Overwhelmed by reactive crisis resolution and inefficient logistics spend. |

## 4. Formal Problem Statement (PS) Matrix

The following formalized analytical table presents the definitive architectural focus points for the engineering phase of the project.

| ID | I am (Customer) | I'm trying to | But | Because | Which makes me feel |
|---|---|---|---|---|---|
| **PS-1** | A regional agricultural extension officer. | Advise clusters of local farmers on optimal crop rotation | Soil data profiles are severely outdated, leading to | No unified digital infrastructure exists to aggregate | Frustrated by the inability to offer high- |

| ID | I am (Customer) | I'm trying to | But | Because | Which makes me feel |
|---|---|---|---|---|---|
| | | and fertilizer usage inputs. | general, ineffective advice. | dynamic laboratory soil logs. | impact, tailored support. |
| **PS-2** | An agribusiness commodity risk trader. | Forecast market prices accurately to balance domestic agricultural procurement contracts. | Spot market price spikes happen without warning, inducing huge structural losses. | Historical pricing streams are siloed separate from current weather anomaly metrics. | Exposed to immense financial risk and market vulnerability. |
| **PS-3** | A smart irrigation systems coordinator. | Automate water distribution controls across drought-prone rural zones. | Severe ground water over-extraction occurs during dry seasonal periods. | Sensor networks operate locally without incorporating predictive monsoon timelines. | Guilty of resource depletion due to rigid baseline mechanics. |